Bayesian AB Testing

MA 03/22/2026

Problem Framing

Given observed conversion rates for landing pages A, B. What is the posterior distribution of each conversion rate? What is the probability B beats A, and what decision would we make under uncertainty?

Observed Data

- Landing page A: 42 conversions out of 400 visitors
- Landing page B: 55 conversions out of 390 visitors

Questions

- What is the posterior conversion rates?
- What is $P(\theta_B > \theta_A \mid \text{data})$?
- What decision does the posterior support?

Closed-Form Beta-Binomial Updating

Model Setup

- prior

$\theta_A \sim \text{Beta}(\alpha, \beta)$

$\theta_B \sim \text{Beta}(\alpha, \beta)$

- likelihood

$y_A \mid \theta_A \sim \text{Binomial}(n_A, \theta_A)$

$y_B \mid \theta_B \sim \text{Binomial}(n_B, \theta_B)$

- posterior

$P(\theta_A) \mid \text{data} \propto p(y_A \mid \theta_A)\, p(\theta_A)$

$P(\theta_B) \mid \text{data} \propto p(y_B \mid \theta_B)\, p(\theta_B)$

- Beta-Binomial conjugacy

$$\theta_A \mid \text{data} \sim \text{Beta}(\alpha + y_A,\ \beta + n_A - y_A)$$

$$\theta_B \mid \text{data} \sim \text{Beta}(\alpha + y_B,\ \beta + n_B - y_B)$$

Sample from both posterior distributions

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Observed data
n_A = 400
y_A = 42

n_B = 390
y_B = 55

# Prior
alpha = 1
beta = 1

# Posterior parameters
alpha_A_post = alpha + y_A
beta_A_post = beta + n_A - y_A

alpha_B_post = alpha + y_B
beta_B_post = beta + n_B - y_B

print("Posterior for A: Beta({}, {})".format(alpha_A_post, beta_A_post))
print("Posterior for B: Beta({}, {})".format(alpha_B_post, beta_B_post))

# drawing samples from the posterior
n_samples = 100_000

theta_A_samples = np.random.beta(alpha_A_post, beta_A_post, size=n_samples)
theta_B_samples = np.random.beta(alpha_B_post, beta_B_post, size=n_samples)
```

```
Posterior for A: Beta(43, 359)
Posterior for B: Beta(56, 336)
```

- theta_A_samples is a large set of plausible conversion rates for A
- theta_B_samples is a large set of plausible conversion rates for B
- together they approximate teh posterior distribution

Estimate Posterior Mean

- Posterior mean is just the average of the posterior samples

In [2]:
```
posterior_mean_A = theta_A_samples.mean()
posterior_mean_B = theta_B_samples.mean()
```

```
print("Posterior mean for A:", posterior_mean_A)
print("Posterior mean for B:", posterior_mean_B)
```

```
Posterior mean for A: 0.10695524928344956
Posterior mean for B: 0.1427934190766608
```

- This is the bayesian estimate of each landing page's conversion rate after combining the prior belief and observed data

Compute a 95% credible interval

- the interval containing 95% of the posterior mass

In [3]:
```
ci_A = np.quantile(theta_A_samples, [0.025, 0.975])
ci_B = np.quantile(theta_B_samples, [0.025, 0.975])

print("95% credible interval for A:", ci_A)
print("95% credible interval for B:", ci_B)
```

```
95% credible interval for A: [0.07861613 0.1387555 ]
95% credible interval for B: [0.11000782 0.17893813]
```

- given the model and the data, there is a 95% posterior probability that the true converstion rate lies in this interval

Probability that B beats A

$\text{Directly estimate } \mathrm{P}(\theta_B > \theta_A \mid \text{ data})$

- this is done by checking how often sampled B values exceed sampled A values

In [4]:
```
prob_B_beats_A = np.mean(theta_B_samples > theta_A_samples)
print("P(B > A | data):"), prob_B_beats_A
```

```
P(B > A | data):
```

Out[4]:
```
(None, np.float64(0.93798))
```

- based on the posterior, B has a 93.8% probability of outperforming A

Posterior Distribution of Lift

- lift is $\theta_B - \theta_A$
- this tells you not just that B is better, but by how much

In [5]:
```
lift_samples = theta_B_samples - theta_A_samples
expected_lift = lift_samples.mean()
lift_ci = np.quantile(lift_samples, [0.025, 0.975])
```

```
print("Expected lift:", expected_lift)
print("95% credible interval for lift:", lift_ci)
```

```
Expected lift: 0.03583816979321123
95% credible interval for lift: [-0.00995964  0.08196964]
```

- positive lift means B likely outperforms A
- negative lift means A likely outperforms B
- if the interval includes 0, uncertainty is still meaningful

Posterior Summaries

In [6]:
```
summary_df = pd.DataFrame({
    "Variant": ["A", "B"],
    "Posterior Mean": [posterior_mean_A, posterior_mean_B],
    "95% CI Lower": [ci_A[0], ci_B[0]],
    "95% CI Upper": [ci_A[1], ci_B[1]],
})

summary_df
```

Out[6]:

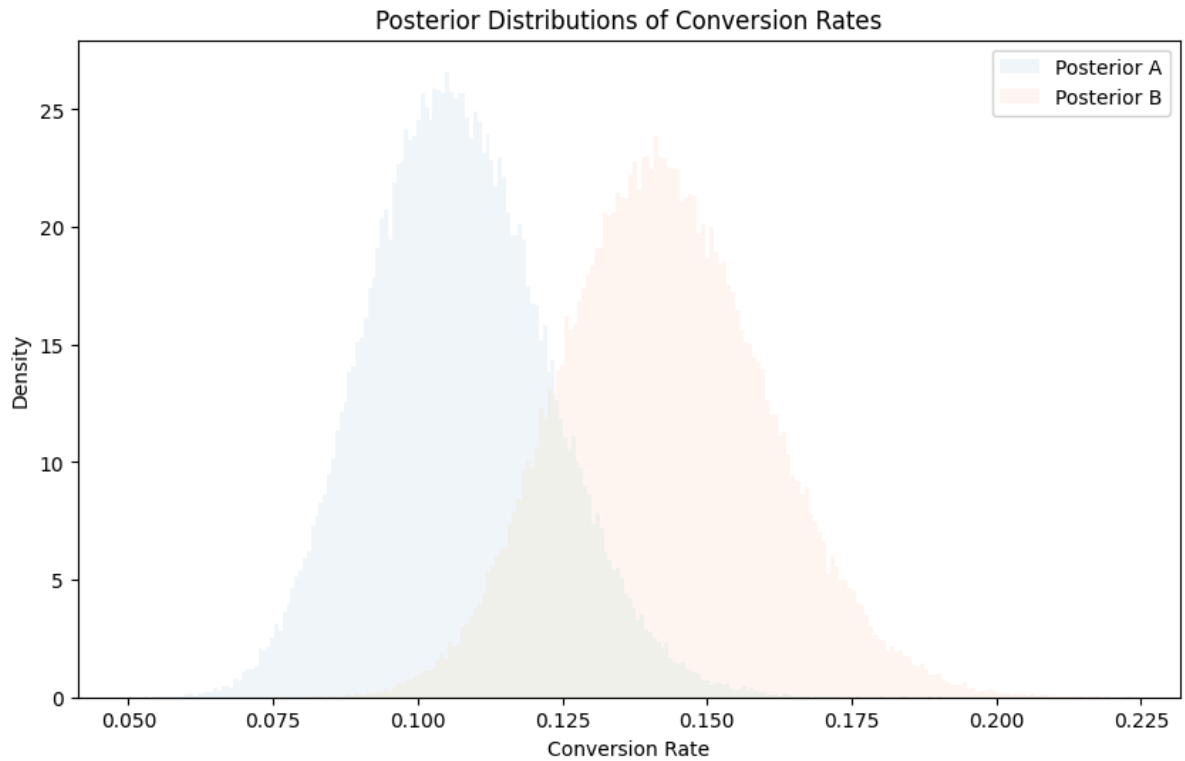
	Variant	Posterior Mean	95% CI Lower	95% CI Upper
0	A	0.106955	0.078616	0.138756
1	B	0.142793	0.110008	0.178938

Visualization

- estimatations to distributions

Posterior density for A and B

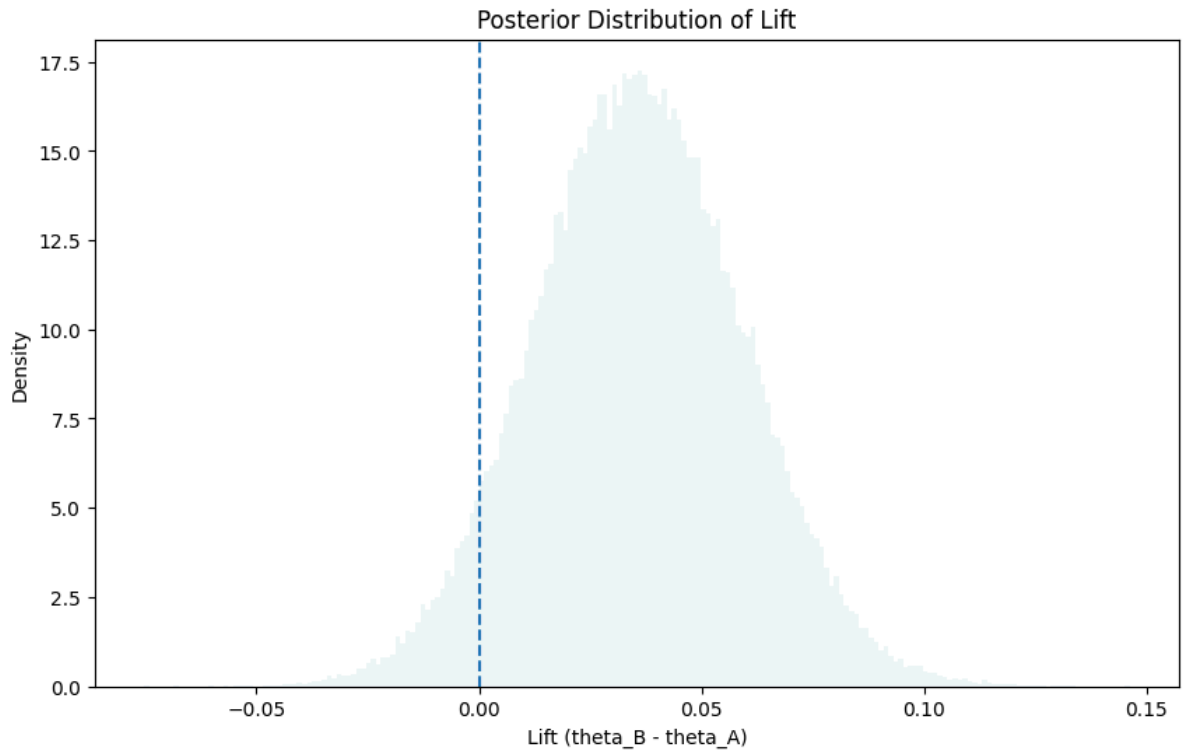
In [7]:
```
plt.figure(figsize=(10,6))
plt.hist(theta_A_samples, bins=200, density=True, alpha=0.05, label="Posteri
plt.hist(theta_B_samples, bins=200, density=True, alpha=0.05, label="Posteri
plt.xlabel("Conversion Rate")
plt.ylabel("Density")
plt.title("Posterior Distributions of Conversion Rates")
plt.legend()
plt.show()
```



- where the plausible values for A and B lie
- how much overlap there is
- whether B is shifted to the right of A

Histogram of Posterior Lift

In [8]:
```
plt.figure(figsize=(10,6))
plt.hist(lift_samples, bins=200, density=True, alpha=0.07)
plt.axvline(0, linestyle="--")
plt.xlabel("Lift (theta_B - theta_A)")
plt.ylabel("Density")
plt.title("Posterior Distribution of Lift")
plt.show()
```



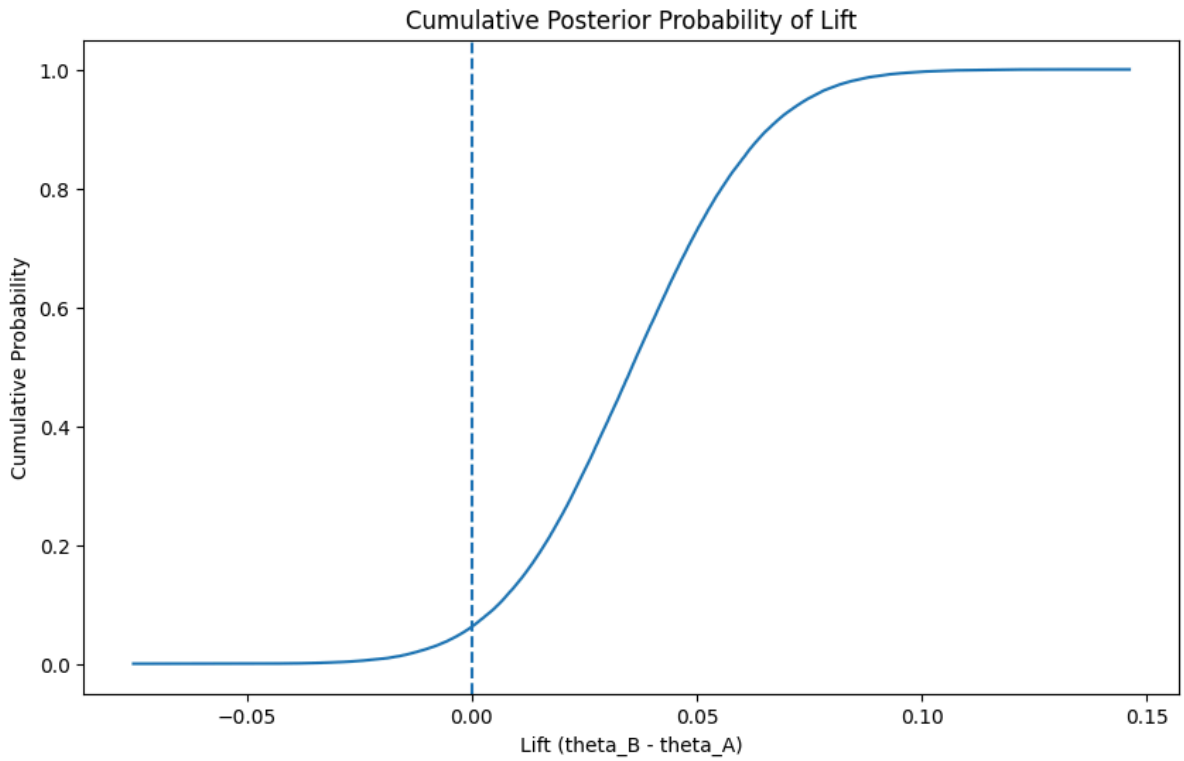
- if most of the lift distribution is above 0, B is likely better
- how large the likely improvement is
- how uncertain that improvement is

Confidence that B beats A

- cumulative curve of lift

In [9]:
```
sorted_lift = np.sort(lift_samples)
cumulative_probs = np.arange(1, len(sorted_lift) + 1) / len(sorted_lift)

plt.figure(figsize=(10,6))
plt.plot(sorted_lift, cumulative_probs)
plt.axvline(0, linestyle="--")
plt.xlabel("Lift (theta_B - theta_A)")
plt.ylabel("Cumulative Probability")
plt.title("Cumulative Posterior Probability of Lift")
plt.show()
```



- the value of the curve at 0 visualizes uncertainty around no improvement

PyMC Version

- how real Bayesian ML work is often done

In [10]:
```
import pymc as pm
import arviz as az

with pm.Model() as ab_model:
    # Priors
    theta_A = pm.Beta("theta_A", alpha=1, beta=1)
    theta_B = pm.Beta("theta_B", alpha=1, beta=1)

    # Likelihoods
    obs_A = pm.Binomial("obs_A", n=n_A, p=theta_A, observed=y_A)
    obs_B = pm.Binomial("obs_B", n=n_B, p=theta_B, observed=y_B)

    # Derived quanitity: lift
    lift = pm.Deterministic("lift", theta_B - theta_A)

    # Sample from posterior
    trace = pm.sample(2000, tune=2000, target_accept=0.95, return_inferencec
```

```
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [theta_A, theta_B]
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 d
raws total) took 1 seconds.
```

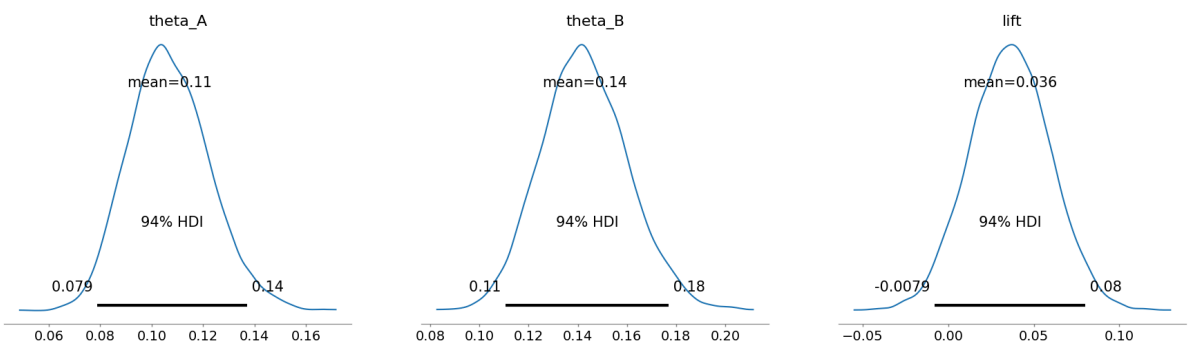
In [11]:
```
az.summary(trace, var_names=["theta_A", "theta_B", "lift"])
```

Out[11]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta_A	0.107	0.016	0.079	0.137	0.0	0.0	4481.0	4358.0	1.0
theta_B	0.143	0.018	0.110	0.177	0.0	0.0	5301.0	4573.0	1.0
lift	0.036	0.024	-0.008	0.080	0.0	0.0	5396.0	4999.0	1.0

- posterior mean, standard deviation, credible intervals, diagnostics, mean, HDI interval, whether lift is mostly positive

In [12]:
```
az.plot_posterior(trace, var_names=["theta_A", "theta_B", "lift"])
plt.show()
```



compute probability B > A for PyMC draws

In [13]:
```
theta_A_draws = trace.posterior["theta_A"].values.flatten()
theta_B_draws = trace.posterior["theta_B"].values.flatten()
lift_draws = trace.posterior["lift"].values.flatten()

prob_B_beats_A_pymc = np.mean(theta_B_draws > theta_A_draws)
print("PyMC estimate of P(theta_B > theta_A | data):", prob_B_beats_A_pymc)
```

```
PyMC estimate of P(theta_B > theta_A | data): 0.93675
```

Decision Interpretation

- what does the prior say?
- how certain are we?
- what action would that support?

In [14]:
```
print("Posterior mean A:", posterior_mean_A)
print("Posterior mean B:", posterior_mean_B)
print("95% CI A:", ci_A)
print("95% CI B:", ci_B)
```

```
print("Probability B > A:", prob_B_beats_A)
print("Expected lift:", expected_lift)
print("95% lift CI:", lift_ci)
```

```
Posterior mean A: 0.10695524928344956
Posterior mean B: 0.1427934190766608
95% CI A: [0.07861613 0.1387555 ]
95% CI B: [0.11000782 0.17893813]
Probability B > A: 0.93798
Expected lift: 0.03583816979321123
95% lift CI: [-0.00995964  0.08196964]
```

The posterior mean converstion rate for B is higher than for variant A. Based on the posterior simulation, the probability that B outperforms A is 0.93735 or 93.735%. The estimated lift, defined as $\theta_B - \theta_A$, is positive on average, suggesting that variant B likely improves conversions. However uncertainty remains, as the CI for the conversion rates still overlap and the posterior lift distribution may include values close to zero. A reasonable decision rule would be to ship variant B only if the posterior probability that it beats A exceeds a threshold such as 90%, or if the expected lift is large enough to justify implementation costs.

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In [ ]: